

A Novel Interleaving Method for High Power Integrated Electric Vehicle Charger with Three-Phase Permanent Magnet Synchronous Motor

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Abstract— The voltage of electric vehicle (EV) batteries has shifted from the conventional 400 V class to the 800 V class, necessitating boost converters to enable charging 800 V class batteries with 400 V class chargers. To reduce the additional costs associated with these boost converters, integrated chargers utilizing a traction motor and inverter have been proposed. Interleaving strategies that employ three-phase windings or only two out of the three phases have been suggested to effectively suppress battery charging current ripple. However, these strategies face challenges such as increased magnetic losses due to interleaving and higher root-mean-square current per phase when conduction is restricted to two phases. These challenges result in temperature rises in permanent magnet synchronous motors, thereby limiting both output and the duration of fast charging. This paper proposes a new interleaving method that uses three-phase windings while suppressing magnetic losses for high-power charging with permanent magnet synchronous motors. The effectiveness of this method is confirmed through finite element analysis and experimental results.

Keywords— *Integrated charger, Step-up converter, DC charging, Electric vehicle, Permanent magnet synchronous motor, magnetic loss reduction, Interleaving*

I. INTRODUCTION

In recent years, there has been a trend toward increasing the capacity of electric vehicle (EV) batteries to extend driving range. Fast charging of high-capacity batteries can be achieved by raising the voltage of EV batteries from the conventional 400 V class to the 800 V class, thereby enhancing the charging output. However, installing a boost converter in EV to charge 800 V class batteries with 400 V class chargers can result in higher equipment costs. To mitigate these challenges without significant cost escalation, integrated chargers that utilize a drive motor and an inverter have been proposed [1]–[10]. The integrated charger can utilize the motor windings, switching

devices of the inverter, smoothing capacitors, harnesses, cooling systems, current sensors, and controller, allowing it to accommodate a high output while maintaining a high cost increases. Nevertheless, the inductance of the motor windings, which are not designed for charging, and the switching frequency of the inverter may not sufficiently suppress the charging current ripple. Charging current ripple can potentially cause overdischarging and overcharging of the battery, which is why standards (IEC 61851-23) specify the allowable limits.

To address the limited inductance of motor windings, strategies such as interleaving with three-phase windings [3]–[5] or using only two of the three phases [1][2] have been suggested to effectively suppress charging current ripple. However, these approaches face challenges, including increased magnetic losses due to interleaving, leading to temperature rises in motor magnets, and heating of motor windings due to higher root-mean-square current per phase when conduction is restricted to two phases. These temperature increases limit both the output and the duration of fast charging. This paper proposes a novel switching method that reduces magnetic losses while energizing all three phases. In previous researches [1]–[5], symmetric phase-shift interleaving has been used, where the switching phases of each UVW phase are evenly shifted according to the number of coils being energized. While symmetric interleaving can minimize the fundamental harmonic component of the charging current ripple, it causes an increase in magnetic losses due to the periodic variation in the direction of the magnetic flux generated by the motor. The proposed method involves designing the phase shifts of interleaving to eliminate the component of magnetic flux which increases the magnetic loss. This paper provides results from finite element analysis (FEA) and experiments that demonstrate the magnetic loss reduction effects of the proposed method.

II. INTERLEAVING FOR MAGNETIC LOSS REDUCTION

This section theoretically explains the reason why applying conventional symmetric interleaving to integrated chargers using motor windings leads to increased magnetic losses. Next, based on the theory, the design method for the phase shift of interleaving to reduce the magnetic loss is proposed.

A. Symmetric phase-shift interleaving of inverter switching

Symmetric interleaving of inverter switching reduces the fundamental frequency ripple in the charging current [1]–[5]. Losses owing to the current ripple generated in the motor intensify because of the d-axis current ripple motor described by a direct-quadrature (dq) coordinate system increases rotor magnet eddy current losses, which warrants attention [11]. The circuit diagram of the three-phase permanent magnet (PM) synchronous motor and inverter is shown in Fig. 1. Each phase of the inverter has complementary upper and lower arm switches, which can be in one of two states: either the upper arm is on or off. Consequently, the three phases of the inverter can adopt one of eight possible switching states, S0–S7, as detailed in Table I. Based on the switching state, the UVW phase voltages v_u , v_v , and v_w take on values of either the inverter voltage E_{dc} or 0. The dq-axes voltages v_d and v_q listed in Table I were obtained via a coordinate transformation expressed as

$$\begin{bmatrix} v_d \\ v_q \end{bmatrix} = \sqrt{\frac{2}{3}} \begin{bmatrix} \cos \theta & \cos\left(\theta - \frac{2}{3}\pi\right) & \cos\left(\theta + \frac{2}{3}\pi\right) \\ -\sin \theta & -\sin\left(\theta - \frac{2}{3}\pi\right) & -\sin\left(\theta + \frac{2}{3}\pi\right) \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} v_u \\ v_v \\ v_w \end{bmatrix} \quad (1)$$

In interleaving switching, when the on-duty D is 50% and the switching phases Φ_u , Φ_v , Φ_w are shifted by one-third of a

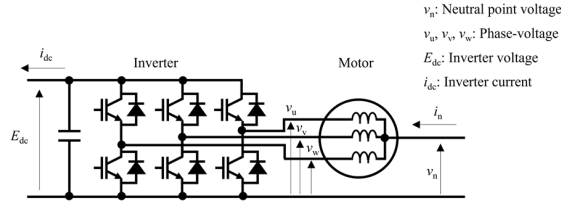


Fig. 1. Schematic of integrated EV charger with traction motor/inverter

switching period, the switching changes at intervals of one-sixth of the period, as depicted in Fig. 2 (a). The relationship between v_d , v_q , and currents i_d and i_q is expressed as follows:

$$\begin{bmatrix} v_d \\ v_q \end{bmatrix} = \begin{bmatrix} R + pL_d & -L_q\omega \\ L_d\omega & R + pL_q \end{bmatrix} \begin{bmatrix} i_d \\ i_q \end{bmatrix} + \begin{bmatrix} 0 \\ \psi_a \end{bmatrix} \quad (2)$$

Here, L_d and L_q denote the inductances of the motor windings along the d-axis and q-axis, respectively. R is the winding resistance, ω denotes the motor's electrical angular velocity, and ψ_a stands for the magnetic flux of the permanent magnet. For instance, with $L_d = 100 \mu\text{H}$, $L_q = 300 \mu\text{H}$, a switching frequency $f_{sw} = 10 \text{ kHz}$, and sufficiently low values of R and ω , the current components i_d and i_q exhibit the pattern shown in Fig. 2(b). In motors where $L_d < L_q$, a relatively large d-axis current ripple can occur, resulting in increased rotor magnetic losses. Because cooling of rotor magnets is difficult, they are prone to demagnetization due to elevated temperatures. Therefore, minimizing rotor magnetic losses is crucial to enhance high-power output.

B. The phase shift design of interleaving for magnetic loss reduction

A novel method for suppressing rotor magnetic losses while interleaving the three phases is proposed in this study. To minimize eddy current losses in the rotor magnet, the d-axis current ripple Δi_d must be reduced to zero. As indicated by (2), maintaining v_d at zero prevents the occurrence of Δi_d . Table I shows that when the rotor angle $\theta = \pi/2$, $v_d = 0 \text{ V}$ in switching states S1 and S4 (in addition to S0 and S7). Therefore, by employing a switching method that alternates between S1 and S4, Δi_d can be eliminated. The transitions between switching states in this method are illustrated in Fig. 3(a). One of the three phases, specifically the u phase, is shifted by half a switching period, making it an interleaving phase where the upper arms of two phases alternate. The i_d and i_q waveforms for this scenario are shown in Fig. 3(b). Similar switching can be achieved by selecting S2 and S5 or S3 and S6 at $\pi/2 + \pi/3$ or $\pi/2 - \pi/3$, respectively, to ensure Δi_d is zero. However, the i_q peak of the proposed interleaving method reaches 36 A, which is 15% higher than that of conventional interleaving. This increase in i_q peak may result in greater motor vibrations and noise; however, the fundamental and harmonic frequencies remain the same or exceed the inverter switching frequency. For instance, a 10 kHz frequency falls within a range unlikely to cause human discomfort.

TABLE I. LIST OF SWITCHING STATES AND VOLTAGES FOR UVW-PHASE AND DQ-AXES

State	U	V	W	v_u	v_v	v_w	v_d	v_q	v_0
S0	0	0	0	0	0	0	0	0	$-\sqrt{3}v_n$
S1	1	0	0	E_{dc}	0	0	$\sqrt{2/3}E_{dc} \cos \theta$	$-\sqrt{2/3}E_{dc} \sin \theta$	$E_{dc}/\sqrt{3} - \sqrt{3}v_n$
S2	1	1	0	E_{dc}	E_{dc}	0	$\sqrt{2/3}E_{dc} \cos(\theta - \pi/3)$	$-\sqrt{2/3}E_{dc} \sin(\theta - \pi/3)$	$2E_{dc}/\sqrt{3} - \sqrt{3}v_n$
S3	0	1	0	0	E_{dc}	0	$-\sqrt{2/3}E_{dc} \cos(\theta + \pi/3)$	$\sqrt{2/3}E_{dc} \sin(\theta + \pi/3)$	$E_{dc}/\sqrt{3} - \sqrt{3}v_n$
S4	0	1	1	0	E_{dc}	E_{dc}	$-\sqrt{2/3}E_{dc} \cos \theta$	$\sqrt{2/3}E_{dc} \sin \theta$	$2E_{dc}/\sqrt{3} - \sqrt{3}v_n$
S5	0	0	1	0	0	E_{dc}	$-\sqrt{2/3}E_{dc} \cos(\theta - \pi/3)$	$\sqrt{2/3}E_{dc} \sin(\theta - \pi/3)$	$E_{dc}/\sqrt{3} - \sqrt{3}v_n$
S6	1	0	1	E_{dc}	0	E_{dc}	$\sqrt{2/3}E_{dc} \cos(\theta + \pi/3)$	$-\sqrt{2/3}E_{dc} \sin(\theta + \pi/3)$	$2E_{dc}/\sqrt{3} - \sqrt{3}v_n$
S7	1	1	1	E_{dc}	E_{dc}	E_{dc}	0	0	$\sqrt{3}(E_{dc} - v_n)$

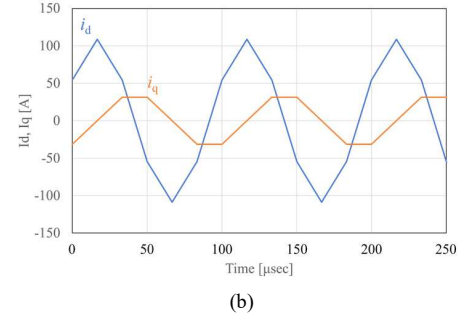
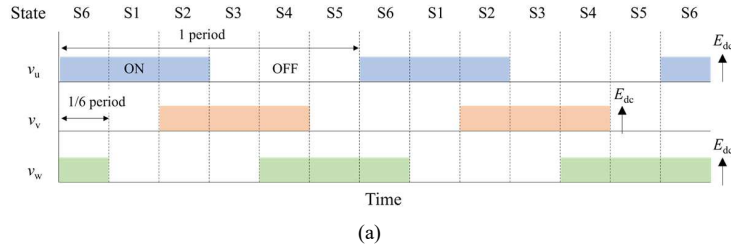


Fig. 2. Conventional interleaving. Duty = 50%, $(\Phi_u, \Phi_v, \Phi_w) = (0^\circ, 120^\circ, 240^\circ)$. (a) Transition of the switching state and phase voltages. (b) dq-axes current.

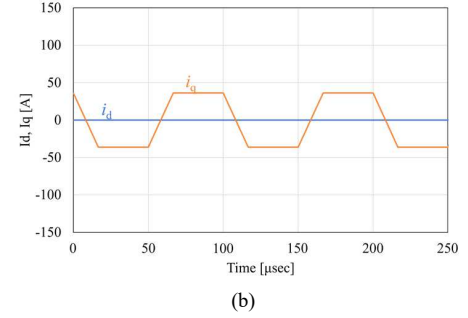
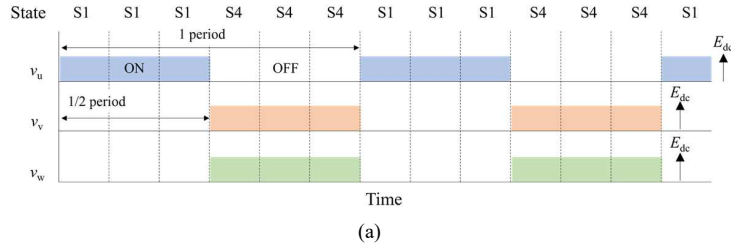


Fig. 3. Proposed interleaving. Duty = 50%, $(\Phi_u, \Phi_v, \Phi_w) = (0^\circ, 180^\circ, 180^\circ)$. (a) Transition of the switching state and phase voltages. (b) dq-axes current.

III. FINITE ELEMENT ANALYSIS OF MOTOR MAGNETIC LOSS

An electromagnetic field analysis using FEA was conducted to verify the effectiveness of the proposed method in reducing magnetic losses. The analysis focused on the drive motor of the Nissan LEAF [12], with the analysis conditions detailed in Table II. The relationship between the rotor electrical angle and the breakdown of iron losses, specifically the stator core loss, rotor core loss, and magnetic loss, when applying both the conventional and proposed methods is illustrated in Fig. 4. At a rotor angle of $\theta = \pi/2$ with the proposed method, magnetic loss is almost completely suppressed, which is expected to prevent output derating due to elevated magnet temperatures. As mentioned in the previous section, by selecting the switching states according to the rotor angle, Δi_d can be reduced to zero with a periodicity of $\pi/3$ electrical angle. Therefore, if the rotor angle during charging can be adjusted by $\pi/3$ electrical angle, the proposed method can effectively always suppress the magnetic loss. In contrast, when examining the magnet losses associated with the conventional method, no clear dependence on the electrical angle is observed. Even at $\theta = 0$, where the magnet loss is minimized, approximately 400 W of magnetic loss still occurs. This indicates that the proposed method effectively reduces magnet losses compared to the conventional method.

Here, it was observed that as the rotor angle approaches $\theta = \pi/2$ with the proposed method, the magnetic losses decrease while rotor core losses tend to increase. The magnetic flux distribution and magnetic loss distribution at a rotor angle of θ

TABLE II. FINITE ELEMENT ANALYSIS CONDITIONS

Parameters	Symbol	Value
Neutral point voltage	v_n	450 V
Inverter voltage	E_{dc}	800 V
Inverter current	i_n	245 A
Switching frequency	f_{sw}	10 kHz
Electrical rotor angle	θ	$0-\pi/2$

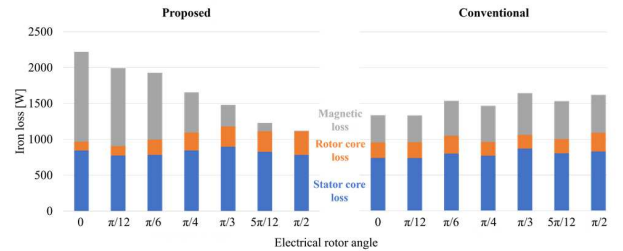


Fig. 4. Breakdown of iron loss analysis for conventional and proposed interleaving methods

= 0 are shown in Fig. 5. the magnetic flux induced from the stator flows perpendicularly into the magnets close to the rotor surface, resulting in the generation of eddy currents. Conversely, at $\theta = \pi/2$, the stator magnetic flux flows into the rotor core, circumventing the rotor magnets, which is the cause of the increased rotor core losses, as shown in Fig. 6.

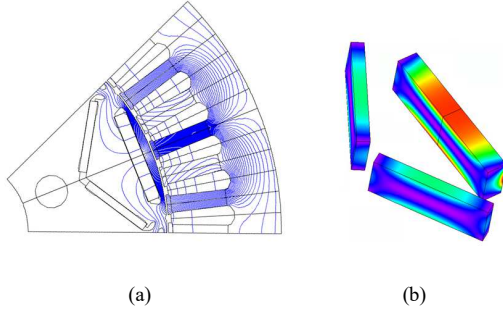


Fig. 5. Rotor angles $\theta = 0$ with the proposed interleaving. (a) Magnetic flux distribution in the motor core and magnet. (b) Loss distribution in the magnet.

IV. EXPERIMENTAL RESULTS

An experiment was conducted to verify the effect of magnetic loss reduction using the proposed method on the temperature rise in an actual EV traction motor. The circuit diagram of the experimental setup is shown in Fig. 7, and the test conditions are detailed in Table III. Owing to facility limitations, a step-down experiment was conducted using v_1 of 400 V with an output limit of 80 kW, alongside v_2 of 320 V. The motor used in the experiment was the drive motor of the Nissan LEAF [12], and the inverter incorporated Si IGBT module (FS600R07A2E3 by Infineon). The IGBTs in the UVW phases were switched with the same duty cycle across all three phases to maintain a neutral-point voltage v_2 of 320 V. The switching phase for the UVW phase is $(\Phi_u, \Phi_v, \Phi_w) = (0^\circ, 120^\circ,$

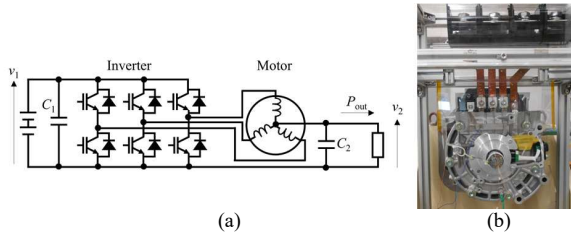


Fig. 7. Experimental bench. (a) Circuit diagram. (b) Test motor.

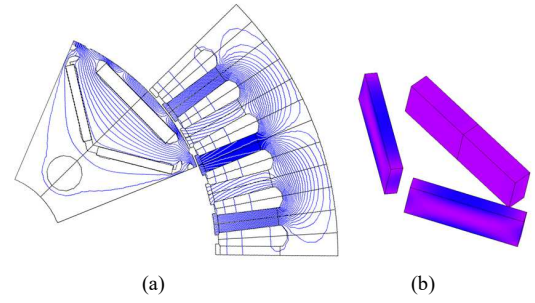
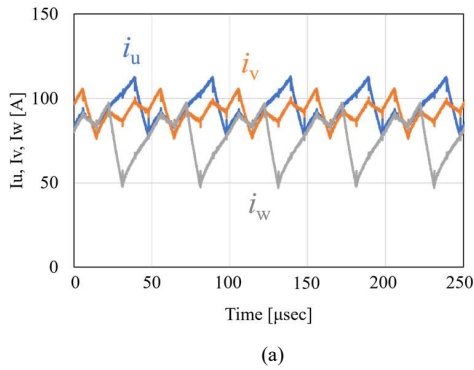


Fig. 6. Rotor angles $\theta = \pi/2$ with the proposed interleaving. (a) Magnetic flux distribution in the motor core and magnet. (b) Loss distribution in the magnet.

$240^\circ)$ in the conventional method, and $(\Phi_u, \Phi_v, \Phi_w) = (0^\circ, 180^\circ, 180^\circ)$ in the proposed method. A continuous load variation test was conducted, with the output increasing stepwise from 5 to 80 kW.

The experimentally obtained waveforms of the UVW phase currents i_u, i_v, i_w and i_d, i_q for both the conventional and proposed interleaving methods are shown in Fig. 8 and Fig. 9, respectively. The proposed control method reduced Δi_d (Fig. 9(b)) by half compared to the conventional three-phase interleaving Δi_d (Fig. 9(a)). The temperature of the rotor magnets, measured using a thermocouple inserted into the rotor magnets via slip rings, confirmed that the temperature rise in the rotor magnets was halved with the proposed control (Fig. 10).

TABLE III. EXPERIMENTAL CONDITIONS

Parameters	Symbol	Value
Inverter voltage	v_1	400 V
Neutral point voltage	v_2	320 V
Switching frequency	f_{sw}	10 kHz
Inverter capacitance	C_1	1277 μ F
Neutral point capacitance	C_2	1277 μ F
Electrical rotor angle	θ	$\pi/2$

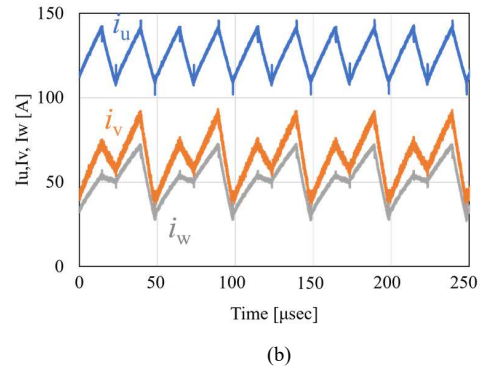


Fig. 8. Waveforms of phase current. (a) Conventional interleaving. (b) Proposed interleaving.

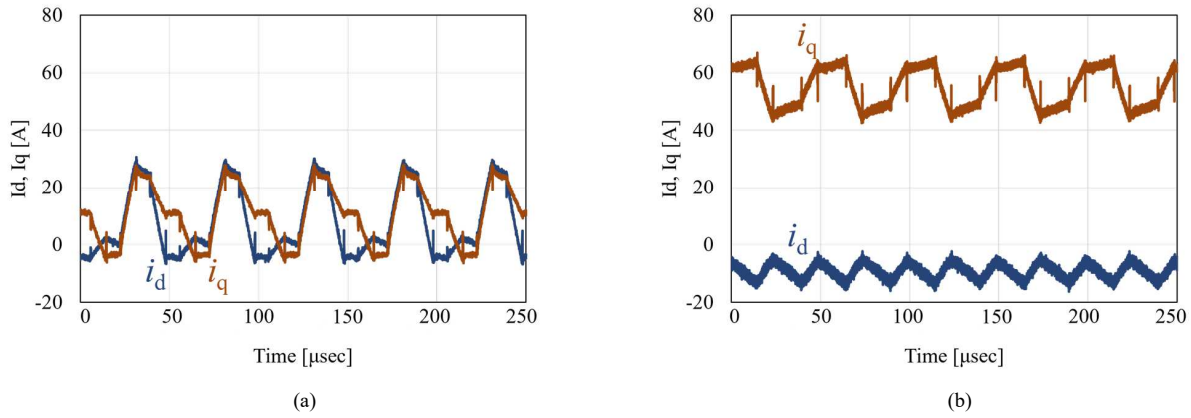


Fig. 9. Waveforms of dq-axes current. (a) Conventional interleaving. (b) Proposed interleaving.

In this experiment, the three-phase current was unbalanced under the proposed control (Fig. 8(b)) due to the open-loop voltage control, leading to relatively lower current values in the V and W phases. Consequently, a comparison of the measurements from the temperature sensor installed in the V-phase coil indicated that the proposed method resulted in a lower coil temperature (Fig. 10). However, if the U-phase coil temperature had been measured, it is likely that a higher temperature would have been observed. Implementing feedback control could improve the balance of the three-phase current. Typically, EV motor drive systems are equipped with current sensors that allow for the measurement of the currents in each UVW phase, which makes it easy to implement feedback control for current balancing.

V. CONCLUSIONS

This study developed a novel interleaving method designed to simultaneously suppress the temperature rise of rotor magnets and motor windings for integrated chargers that utilize a drive motor and inverter. By mitigating the temperature rise, fast charging becomes feasible even after high-speed driving, when the temperatures of the magnets and windings approach their upper limits. The effectiveness of this method was validated through both FEA and experimental testing. The FEA results indicated that magnetic losses generated by conventional three-phase interleaving can be reduced to nearly zero. Additionally, the experimental results demonstrated that the temperature increase in the magnets could be halved. When implementing an unbalanced three-phase interleaving method like the one proposed, incorporating current-balance control for the three phases becomes even more critical.

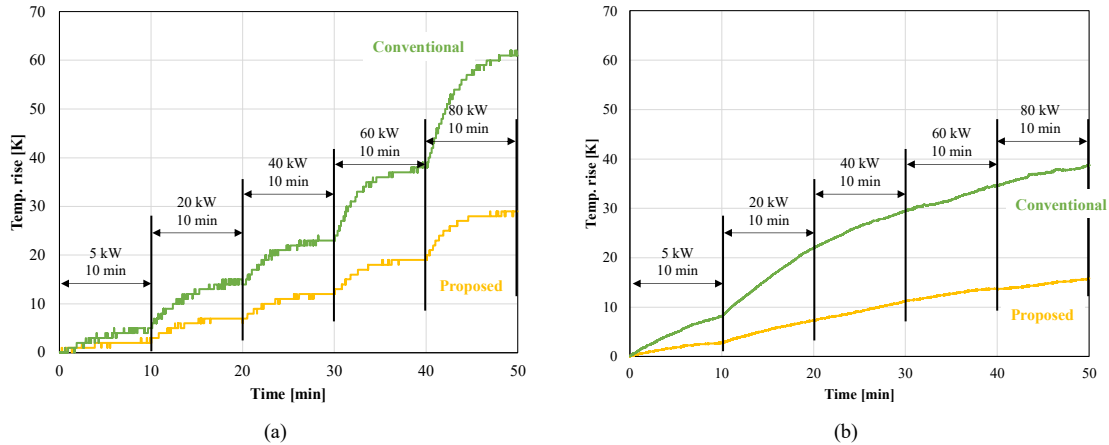


Fig. 10. Transition of motor temperature for conventional and proposed interleaving. (a) Rotor magnet temperature. (b) Stator coil (V-phase) temperature.

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